

Pranav Viswanathan

+1 8573976310 | pvvisthn@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Northeastern University

Boston, MA

Master of Science, Computer Science

Sep. 2024 – May 2026

- Coursework: Machine Learning, MLOps, Algorithms, Foundations of AI, DBMS, Natural Language Processing, Building Scalable Distributed Systems

Vellore Institute of Technology

Chennai, India

Bachelor of Technology, Computer Science and Engineering

2024

- Coursework: Operating Systems, Distributed Systems, Comms for Cyber Physical Systems, Sensors and Actuators

EXPERIENCE

NergyLive

May 2023 – Jul. 2023

Software Development Intern

- Built real-time HVAC anomaly detection using isolation forests and statistical process control, processing 10K+ sensor readings/minute with sub-100ms latency in Python
- Built interactive Plotly dashboards with drill-down capabilities for 20+ system parameters, enabling engineers to identify root causes 3x faster and reducing average troubleshooting time from 45 to 15 minutes
- Deployed anomaly detection system into a live production environment in collaboration with cross-functional teams, reducing downtime incidents and cutting maintenance costs by 15%

PROJECTS

SwarmCoordinator: Distributed Drone Swarm System | *Python, FastAPI, Three.js, Next.js* Jan. 2026 – Feb. 2026

- Implemented Raft consensus algorithm from scratch across 10 simulated drone agents for distributed leader election and fault tolerance, supporting dynamic term changes and candidate promotion with configurable 10% packet loss simulation
- Engineered physics-based simulation engine using Artificial Potential Fields for real-time collision avoidance and flocking behavior, with dynamic formation control (grid, circle, random) streamed over WebSocket at 10Hz to a Three.js 3D visualization
- Built full-stack system with FastAPI REST backend exposing 7+ endpoints for drone management, mission control, and consensus monitoring, paired with a Next.js frontend and automated shell scripts for one-command deployment

Lucera: Advanced Video Processing Pipeline | *Python, Docker, FFmpeg, PyTorch* Dec. 2025 – Jan. 2026

- Engineered modular Python video processing pipeline integrating Real-ESRGAN 4x upscaling, RIFE frame interpolation (24fps to 60fps), and Whisper AI captioning with VMAF/PSNR quality metrics above industry-standard threshold of 75 across 6 autonomous stages
- Architected loosely-coupled OOP classes orchestrating FFmpeg subprocess management, GPU batch inference, and adaptive HLS streaming with 4 bitrate variants, achieving sub-200ms frame analysis latency validated through 40+ pytest unit tests
- Containerized end-to-end pipeline using Docker with automated shell scripts for one-command deployment, enabling persistent artifact storage across denoising, interpolation, upscaling, and packaging stages

Rebalance-AI: Bluebikes Demand Prediction | *Python, Airflow, MLflow, Docker, XGBoost* Sep. 2025 – Dec. 2025

- Engineered end-to-end ML pipeline processing 3.16M ride records with Apache Airflow orchestration, implementing automated feature engineering for weather, temporal, and geospatial data
- Developed and deployed production-grade ensemble models (XGBoost, LightGBM, Random Forest) achieving $R^2 > 0.96$, with MLflow for experiment tracking, model versioning, and deployment automation
- Built CI/CD pipeline using GitHub Actions for automated testing and Docker containerization, ensuring reproducible model training and deployment across 400+ bike stations

TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, C/C++, JavaScript, R, MATLAB

ML & Data Science: PyTorch, Scikit-learn, XGBoost, LightGBM, OpenCV, Hugging Face, Pandas, NumPy, Matplotlib, Seaborn

MLOps & Infrastructure: Apache Airflow, MLflow, Docker, GitHub Actions, Terraform, AWS

Concepts: Supervised/Unsupervised Learning, Reinforcement Learning, Feature Engineering, Distributed Systems, Big Data Processing

Certifications: Google Cloud Essentials, Create and Manage Google Cloud Resources, Postman Student Expert, Supervised ML: Regression and Classification